

Consider the optimization problem

$$\min -3x_1 - 4x_2$$

subject to

$$-x_1 + x_2 \leq 2$$

$$-3x_1 + 2x_2 \geq -5$$

$$x_1 + x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

1. Solve the problem by calculating the value of the objective function at each vertex.
2. Write the dual of this problem.
3. Use the complementarity slackness conditions to solve the dual.